Advanced Programming Practice · 21CSC203P

Unit 4 · Session 2 · SLO 1 — Solved Worksheet

Name: DEEP SAND DEEP SAND | **Reg. No:** RA2511026010557

Unit â€“ 4 : Session â€“ 2 : SLO â€“ 1 | Pure Functions with Examples

I. Match the Following

1. map() â€” Applies a function to each element. Example: list(map(str.upper, ["a", "b"])) â† ["A", "B"].
2. filter() â€” Selects elements based on a condition. Example: list(filter(lambda x: x > 0, [-1, 2, -3, 4])) â† [2, 4].
3. reduce() â€” Accumulates values to a single result. Example: reduce(lambda a, b: a + b, [1, 2, 3, 4]) â† 10.
4. Pure Function â€” Same output for the same input, with no side effects. Example: def add(a, b): return a + b.

II. Key Notes

• A pure function never mutates its arguments, global variables, or performs I/O â€” its result depends only on its inputs.
• map() and filter() return lazy iterators in Python 3; wrap them in list() to materialize the results.
• reduce() lives in the functools module: from functools import reduce.